



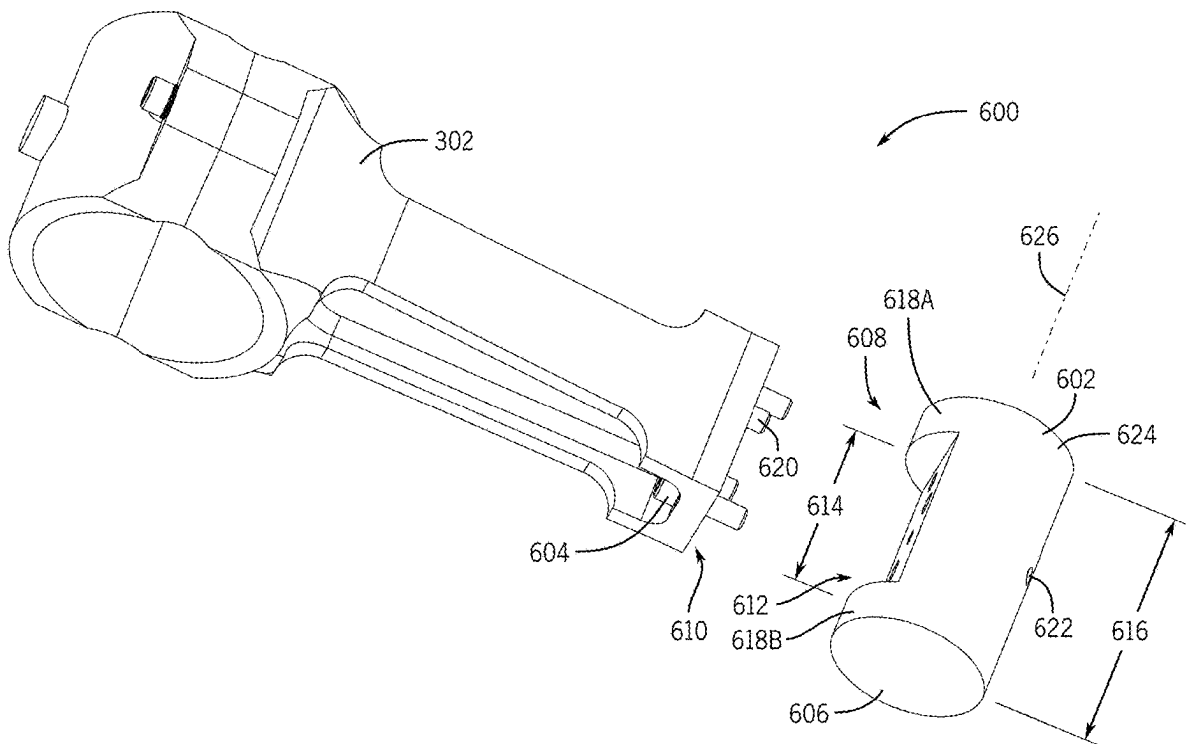
US 20250277487A1

(19) **United States**(12) **Patent Application Publication**
Ellisor et al.(10) **Pub. No.: US 2025/0277487 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **CONNECTING ROD ASSEMBLY FOR
RECIPROCATING PUMP***F04B 53/18* (2006.01)*F16C 5/00* (2006.01)(71) Applicant: **Vulcan Industrial Holdings, LLC**,
Houston, TX (US)(52) **U.S. Cl.**CPC *F04B 53/147* (2013.01); *F04B 19/22*
(2013.01); *F04B 53/18* (2013.01); *F16C 5/00*
(2013.01)(72) Inventors: **Kyle Matthew Ellisor**, Houston, TX
(US); **Chance Ray Mullins**, Houston,
TX (US)(21) Appl. No.: **19/093,532**

(57)

ABSTRACT(22) Filed: **Mar. 28, 2025****Related U.S. Application Data**(63) Continuation of application No. 17/458,754, filed on
Aug. 27, 2021.(60) Provisional application No. 63/071,137, filed on Aug.
27, 2020.**Publication Classification**(51) **Int. Cl.***F04B 53/14* (2006.01)*F04B 19/22* (2006.01)

A connecting rod assembly includes a crosshead for coupling a connecting rod to a plunger, the crosshead having a hole extending through the crosshead. The connecting rod assembly also includes a knuckle cage positioned within the crosshead, the knuckle cage extending through the hole and including a cutout to provide access to a passage formed in the knuckle cage. The connecting rod assembly further includes a knuckle positioned within the passage of the knuckle cage, the knuckle having a mounting surface that is substantially planar, the mounting surface coupling to the connecting rod at a mating end with a substantially planar mating end surface.



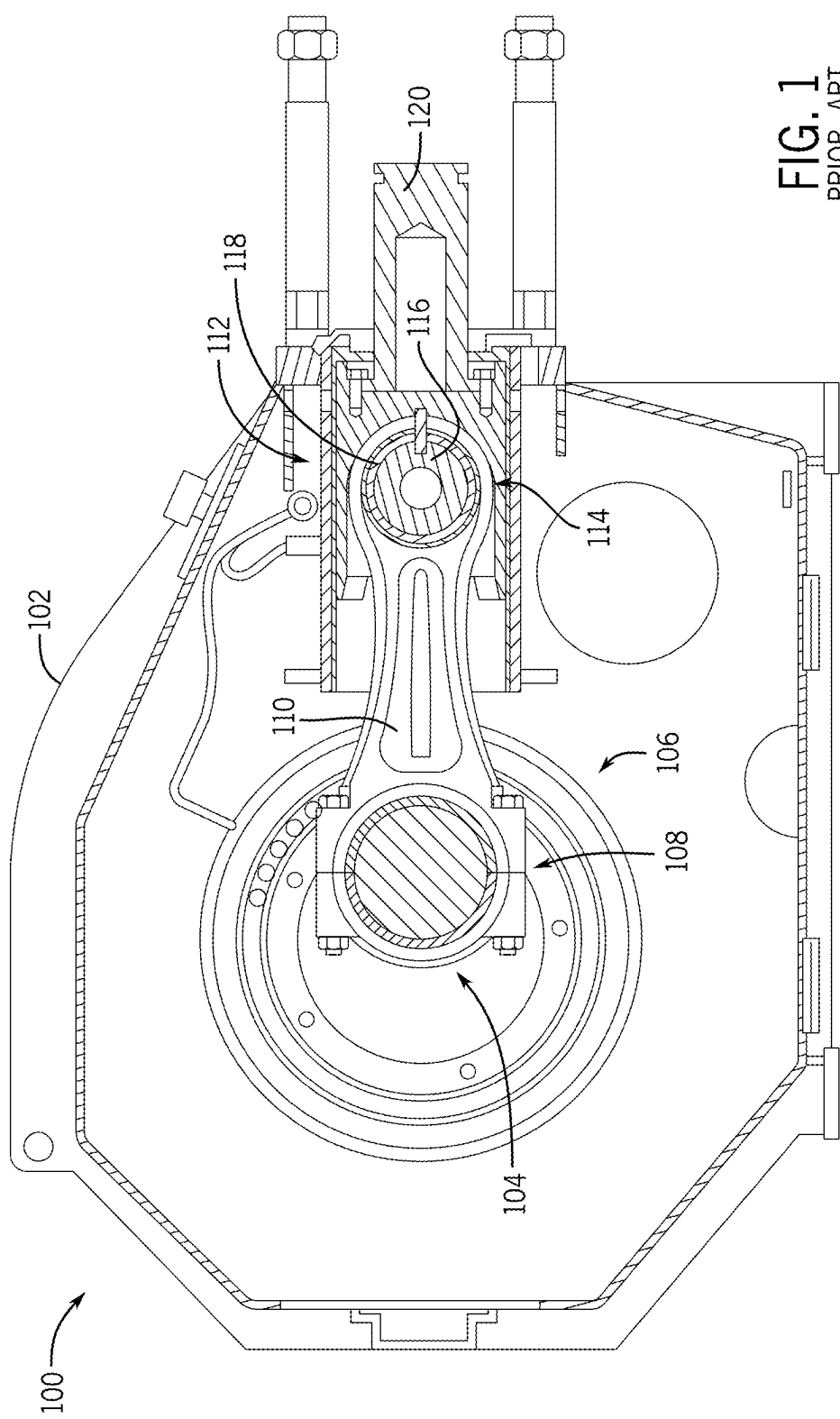


FIG. 1
PRIOR ART

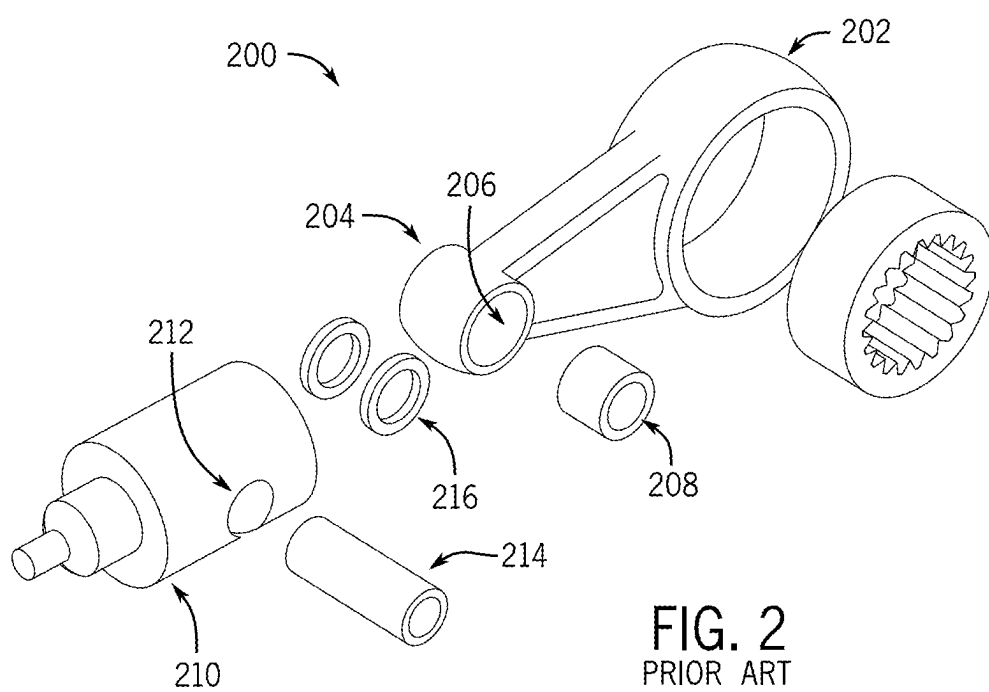


FIG. 2
PRIOR ART

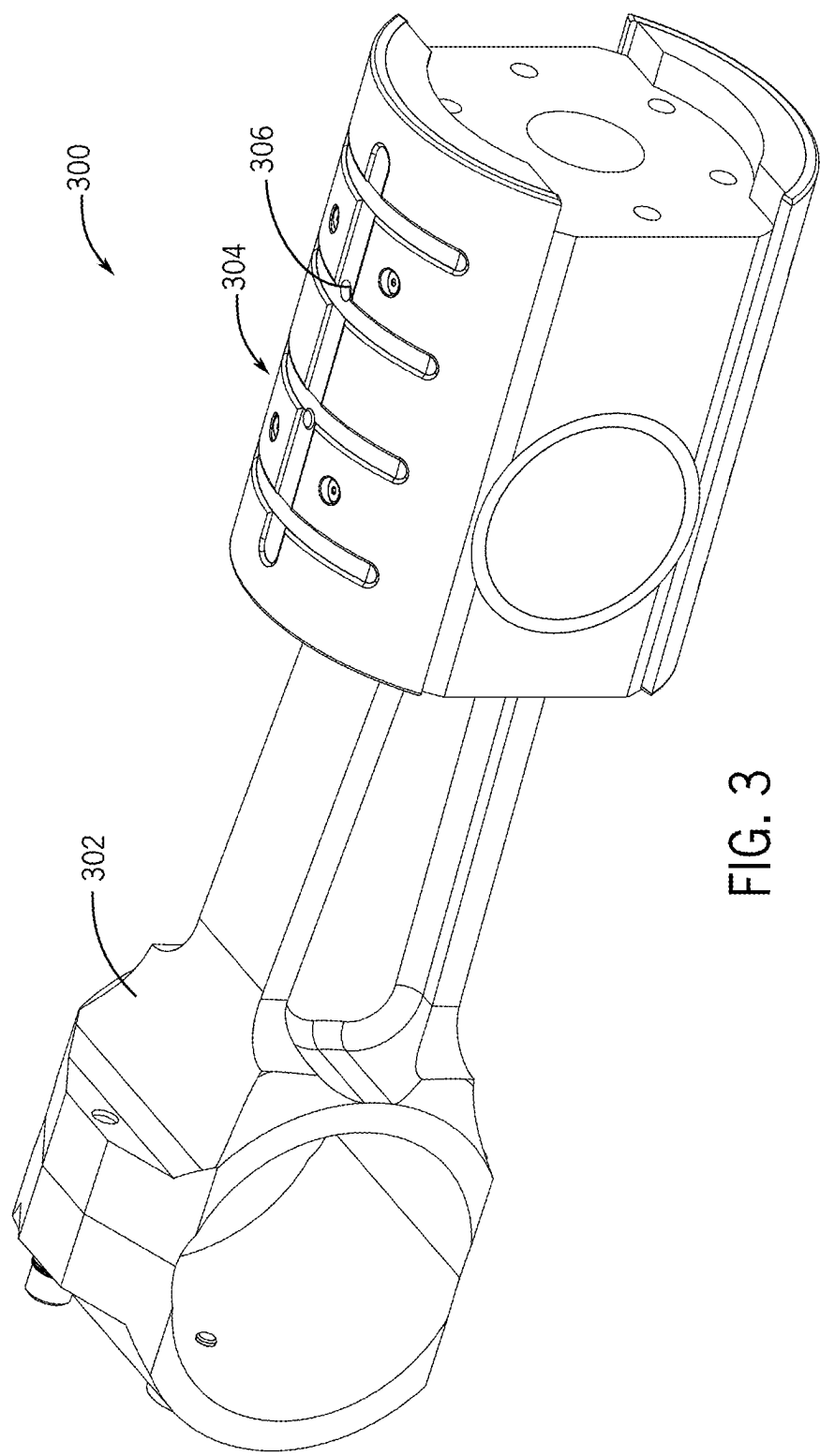
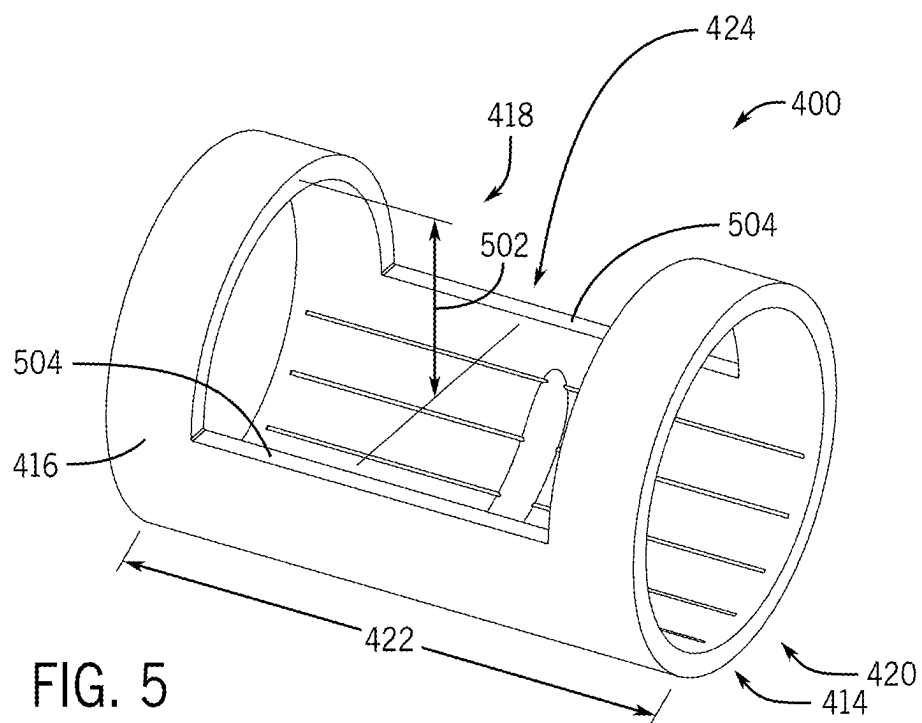
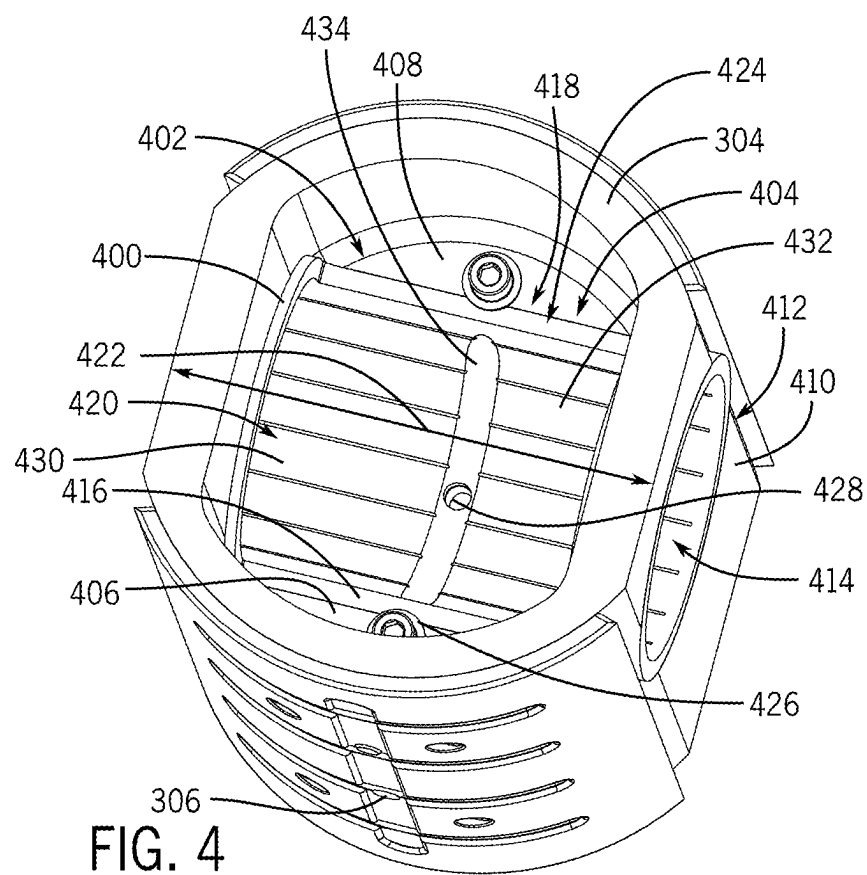
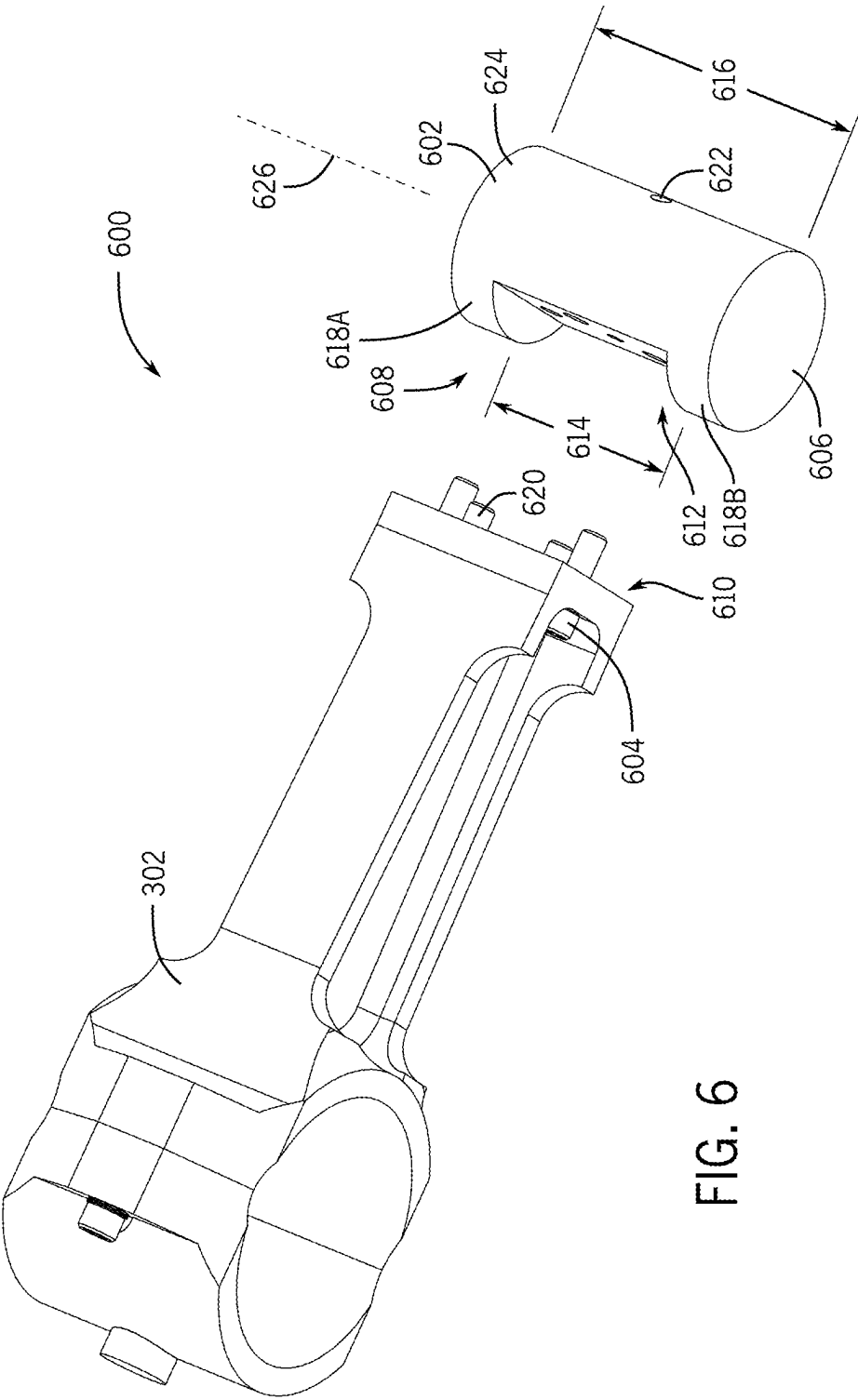
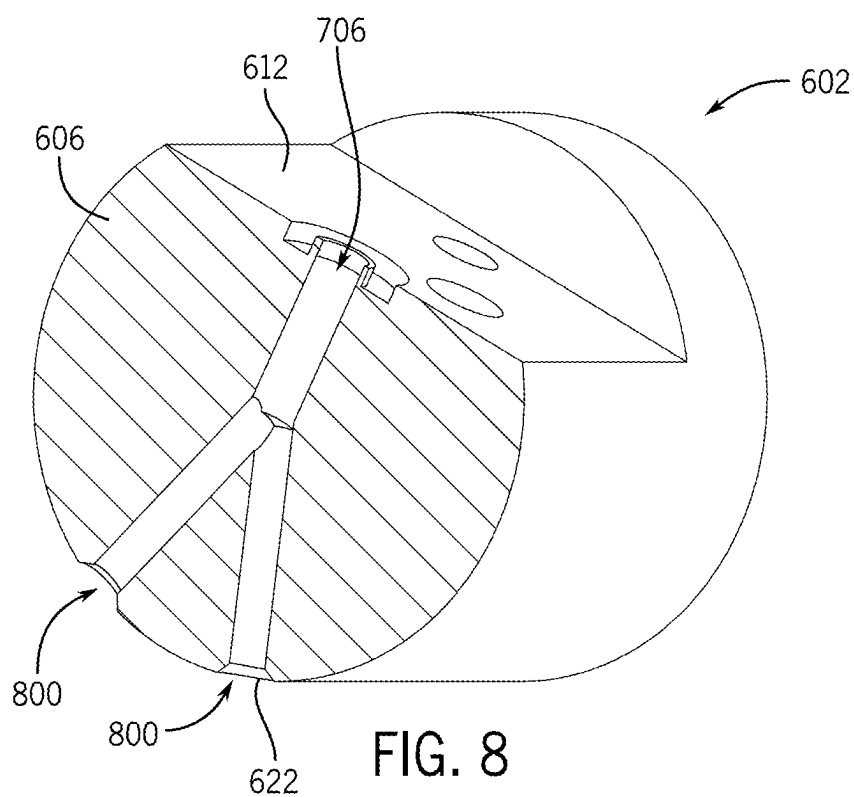
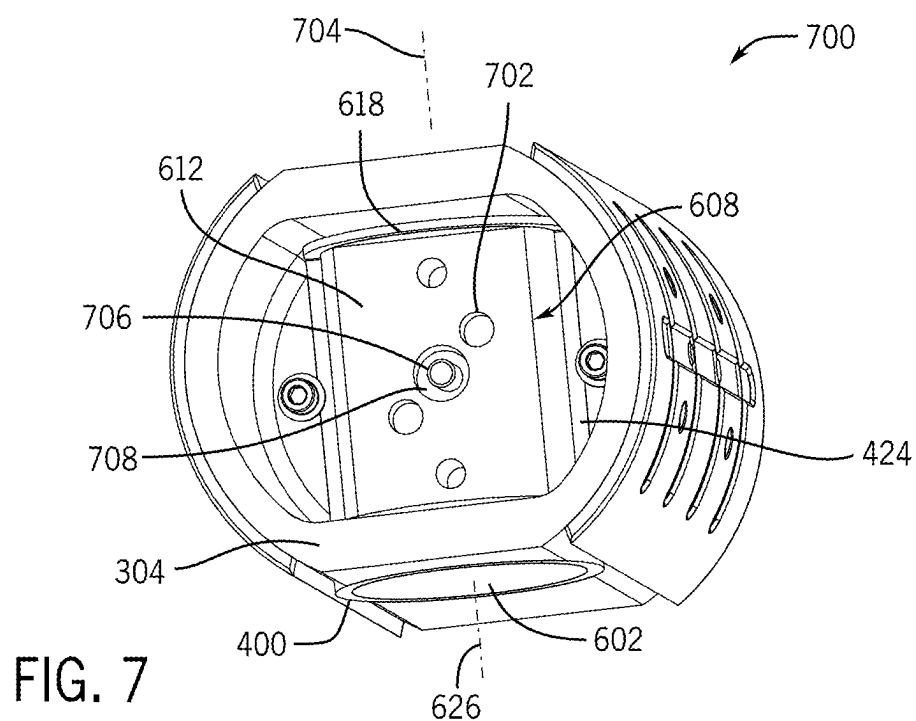


FIG. 3







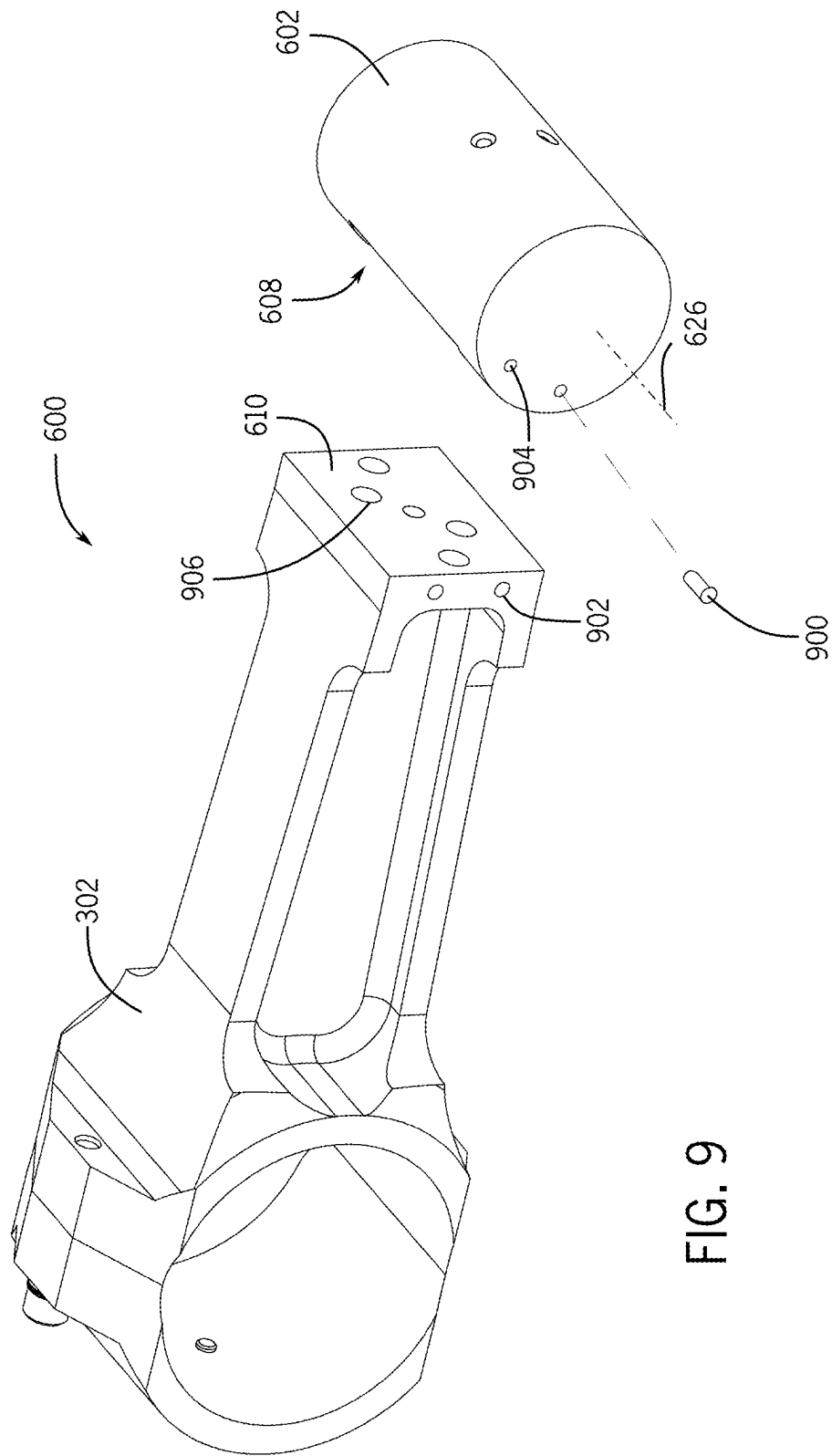


FIG. 9

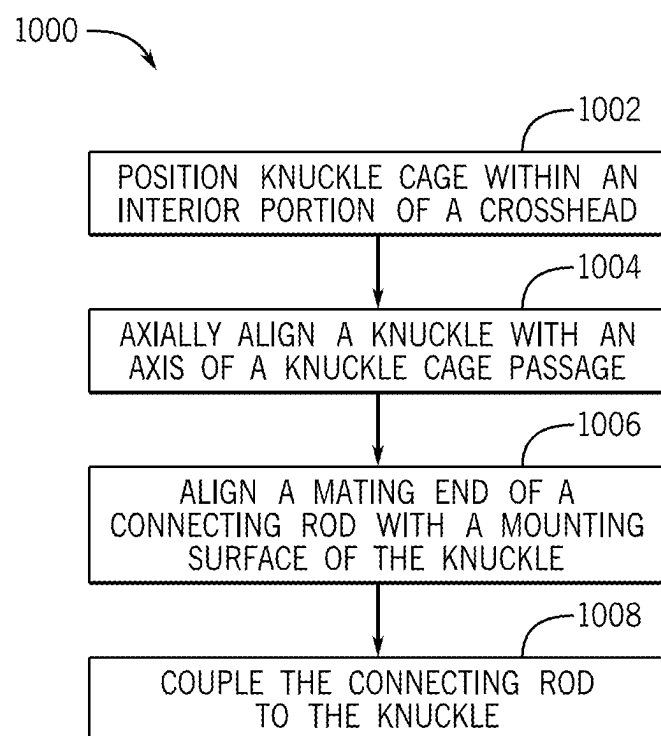


FIG. 10

CONNECTING ROD ASSEMBLY FOR RECIPROCATING PUMP

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. Non-Provisional application Ser. No. 17/458,754, filed Aug. 27, 2021, titled “CONNECTING ROD ASSEMBLY FOR RECIPROCATING PUMP,” which claims the benefit of U.S. Provisional Patent Application No. 63/071,137, filed Aug. 27, 2020, titled “CONNECTING ROD ASSEMBLY FOR RECIPROCATING PUMP,” the full disclosures of which are hereby incorporated herein in their entireties by reference.

TECHNICAL FIELD

[0002] Embodiments of the subject matter disclosed herein generally relate to pump systems, and in particular to connecting rod and crosshead assemblies.

BACKGROUND

[0003] Pumping systems may be used in a variety of applications, such as industrial applications where pumping systems are used to elevate a working fluid pressure. One such application is hydraulic fracturing systems, where high pressure pumps are used to increase a fluid pressure of a working fluid (e.g., fracturing fluid, slurry, etc.) for injection into an underground formation. The working fluid may include particulates, which are injected into fissures of the formation. When the fluid is removed from the formation, the particulates remain and “prop” open the fissures, facilitating flow of oil and gas. In many applications, reciprocating pumps are used where a fluid is introduced into a fluid end inlet passage and out through an outlet passage. A plunger reciprocates within a bore to add energy to the fluid. The plunger may be coupled to a powered crankshaft that provides energy via a connecting rod. The connecting rod is coupled to the plunger via a crosshead. The end connection at the crosshead typically has a bushing located within the crosshead. However, loading at the bushing may be uneven.

SUMMARY

[0004] Applicant recognized the problems noted above herein and conceived and developed embodiments of systems and methods, according to the present disclosure, for crosshead assemblies.

[0005] In accordance with one or more embodiments, a connecting rod assembly includes a crosshead for coupling a connecting rod to a plunger, the crosshead having a hole extending through the crosshead. The connecting rod assembly also includes a knuckle cage positioned within the crosshead, the knuckle cage extending through the hole and including a cutout to provide access to a passage formed in the knuckle cage. The connecting rod assembly further includes a knuckle positioned within the passage of the knuckle cage, the knuckle having a mounting surface that is substantially planar, the mounting surface coupling to the connecting rod at a mating end with a substantially planar mating end surface.

[0006] In accordance with another embodiment, a connecting rod assembly includes a crosshead having an interior portion, a hole extending from a first end to a second end, and a pair of platforms on opposite first and second sides of

the hole. The connecting rod assembly also includes a knuckle cage positioned within the hole and extending from the first end to the second end, a cage axis being substantially perpendicular to the pair of platforms, and a recess formed in the knuckle cage extending through a body portion of the knuckle cage to provide access to a pivot surface of the knuckle cage from a direction perpendicular to the cage axis, the pivot surface forming at least a portion of a cage passage extending along the cage axis. The connecting rod assembly further includes a knuckle positioned within the cage passage, the knuckle being axially aligned with the cage axis such that the knuckle, responsive to a force, rotates about the cage passage, the knuckle having a planar mating surface positioned to align with the recess formed in the knuckle cage.

[0007] In accordance with another embodiment, a method includes positioning a knuckle cage within a crosshead. The method also includes positioning a knuckle within the knuckle cage. The method further includes aligning a mating end of a connecting rod with a mounting surface of the knuckle. The method also includes coupling the connecting rod to the knuckle.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The present technology will be better understood on reading the following detailed description of non-limiting embodiments thereof, and on examining the accompanying drawings, in which:

[0009] FIG. 1 is a schematic cross-sectional view of a prior art pump assembly;

[0010] FIG. 2 is an isometric view of a prior art connecting rod assembly;

[0011] FIG. 3 is an isometric view of an embodiment of an assembled configuration of a connecting rod and crosshead, in accordance with embodiments of the present disclosure;

[0012] FIG. 4 is an isometric view of an embodiment of a crosshead with a knuckle cage, in accordance with embodiments of the present disclosure;

[0013] FIG. 5 is an isometric view of an embodiment of a knuckle cage, in accordance with embodiments of the present disclosure;

[0014] FIG. 6 is an isometric view of an embodiment of a connecting assembly, in accordance with embodiments of the present disclosure;

[0015] FIG. 7 is an isometric view of an embodiment of a crosshead assembly, in accordance with embodiments of the present disclosure;

[0016] FIG. 8 is an isometric view of an embodiment of a knuckle, in accordance with embodiments of the present disclosure;

[0017] FIG. 9 is an isometric view of an embodiment of a connecting assembly, in accordance with embodiments of the present disclosure; and

[0018] FIG. 10 is a flow chart of an embodiment of a method for coupling a connecting rod to a knuckle, in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0019] The foregoing aspects, features, and advantages of the present disclosure will be further appreciated when considered with reference to the following description of embodiments and accompanying drawings. In describing the embodiments of the disclosure illustrated in the appended

drawings, specific terminology will be used for the sake of clarity. However, the disclosure is not intended to be limited to the specific terms used, and it is to be understood that each specific term includes equivalents that operate in a similar manner to accomplish a similar purpose.

[0020] When introducing elements of various embodiments of the present disclosure, the articles “a”, “an”, “the”, and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including”, and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Any examples of operating parameters and/or environmental conditions are not exclusive of other parameters/conditions of the disclosed embodiments. Additionally, it should be understood that references to “one embodiment”, “an embodiment”, “certain embodiments”, or “other embodiments” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Furthermore, reference to terms such as “above”, “below”, “upper”, “lower”, “side”, “front”, “back”, or other terms regarding orientation or direction are made with reference to the illustrated embodiments and are not intended to be limiting or exclude other orientations or directions. Additionally, like reference numerals may be used for like components, but such use is for convenience purposes and not intended to limit the scope of the present disclosure. Moreover, use of terms such as substantially or approximately may refer to ± 10 percent.

[0021] Embodiments of the present disclosure are directed toward a connecting rod assembly for coupling a plunger for a pump (e.g., a reciprocating pump) to a power end. As will be described herein, multiple configurations may include knuckles and knuckle cages, arranged within a crosshead, to enable different mounting configurations between the crosshead and the connecting rod.

[0022] In embodiments, the crosshead is loaded with a knuckle cage, which is then prevented from rotating by one or more keys, such as a pair of locking washers, among other options. The retention may or may not also include a feature to prevent axial motion of the knuckle cage relative to the cross-bored hole of the crosshead. The knuckle is then inserted from the side of the crosshead into the knuckle cage and orientated such that a flat section is pointed to a rear of the crosshead. The connecting rod is then attached to knuckle with a pair of axially aligned fasteners that may not experience any bending or shear loading due to their configuration. There may also be additional alignment pins, keys, or bosses in various embodiments.

[0023] Various embodiments may also include the connecting rod assembly where the knuckle and connecting rod are attached via fasteners/pins/keys that are slid through the side of the parts. This could also include features such as a T-slot or dovetail.

[0024] As will be described herein, a reciprocating pump assembly has a pump within a housing. The pump has a crankshaft that rotates and a crosshead that is connected to a plunger via a pony rod. The plunger pumps fluid through cylinders in the pump. A connecting rod connects at one end to the crankshaft and at another end to the crosshead to translate the rotational motion of the crankshaft into linear movement of the crosshead, which is translated to the plunger. The end connected to the crosshead is secured to the crosshead traditionally by a bushing located within the

crosshead that allows the connecting rod end to pivot during operation. Embodiments of the present disclosure include an independent knuckle piece to make the assembly. The knuckle may be a smaller part that may be hardened independently of the connecting rod casting. The knuckle may be attached via pins and fasteners to the connecting rod such that it provides the pivot point of the assembly against a bearing. The interface between the knuckle pin and the connecting rod may be a flat-to-flat surface where the loading is distributed evenly, in various embodiments.

[0025] Embodiments of the present disclosure provide for a distinct coupling between the connecting rod and the crosshead. As will be described, the traditional pin connection is removed, which moves expensive heat treating and carburizing processes into a single smaller component, where the need for additional machining post hardening would not be necessary. Additionally, the number of load bearing components is significantly reduced from a typical assembly. Accordingly, embodiments may be directed toward a flat-to-flat knuckle pin and connecting rod interface and side assembly, rather than a traditional axial assembly.

[0026] FIG. 1 is a schematic cross-sectional view of an embodiment of a pump assembly 100, which may also be referred to as a reciprocating pump assembly and/or a reciprocating pump. The pump assembly 100 may be utilized during hydraulic fracturing operations, among other operations, where a working fluid (e.g., fracturing fluid, slurry, etc.) is introduced into the fluid end and energy is added to the working fluid to increase a pressure of the working fluid. Fracturing fluid, by way of example only, may include corrosives and also particulates, such as sand or ceramics, which are utilized during fracturing operations.

[0027] The illustrated pump assembly 100 includes a frame or housing 102 that contains a powered crankshaft 104 that rotates about an axis (not pictured, but perpendicular to the plane of the page). It should be appreciated that various components have been removed for clarity and conciseness. The crankshaft 104 illustrated in FIG. 1 is coupled to a connecting rod 106 at a first end 108. The connecting rod 106 includes a body 110 that extends laterally away from the powered crankshaft 104 and into a crosshead 112. A second end 114 of the connecting rod 106 is positioned within the crosshead 112 and may be coupled via a pin 116 that extends through a bearing 118 positioned through the second end 114. The crosshead 112 is further coupled to a pony rod 120 that transmits energy from the powered crankshaft 104 into a plunger (not pictured). In operation, rotational movement of the powered crankshaft 104 is transferred to the connecting rod 106, which may pivot about the bearing 118 within the crosshead 112 to translate force from the powered crankshaft 104 into axial movement applied to the plunger (not pictured).

[0028] FIG. 2 is a perspective exploded view of a prior art connecting rod assembly 200. A connecting rod 202 is illustrated with an end 204 and an opening 206 that receives a journal bearing 208. The end 204 is installed within a crosshead 210 such that a radial hole 212 of the crosshead 210 aligns with the journal bearing 208. A pin 214 may extend through the radial hole 212 and the journal bearing 208, with bushings 216 positioned on the outside of the pin 214. Accordingly, loading may be either entirely through the connecting rod 202 and into the crosshead 210 and/or through the connecting rod 202, journal bearing 208, pin 214, and bushings 216. Embodiments of the present disclo-

sure may overcome the problems associated with this traditional configuration, which may lead to expensive treatment of various components to resist wear from loading.

[0029] It should be appreciated that alternative mounting configurations than those shown in FIG. 2 may also be used that have similar drawbacks. For example, a rounded interface between the crosshead pin and crosshead, for example where a fastener may be used to secure the pin, may present alignment problems. Additionally, shear loading may be applied during a retract stroke. Moreover, such a configuration provides a small area on the pin/crosshead interface, which could lead to assembly difficulties, among other problems. Embodiments of the present disclosure address and overcome these problems.

[0030] FIG. 3 is an isometric view of an embodiment of an assembled configuration **300** including a connecting rod **302** and a crosshead **304**. As will be described below, embodiments may include a removable knuckle and knuckle cage within the crosshead **304** to facilitate improved coupling between the components. For example, the knuckle may enable a flat-to-flat mounting configuration with the connecting rod **302**. Additionally, in various embodiments, coupling between the knuckle and the connecting rod **302** may be axial (e.g., via axial fasteners) to reduce loading, such as shear loads, on the fasteners. Furthermore, various other coupling methods such as radial pins, J-slots, T-slots, dovetails, and the like may also be utilized with embodiments of the present disclosure. Moreover, one or more apertures **306** may be utilized to provide lubrication to interior components, thereby providing a hydrodynamic cushion to the knuckle and/or a knuckle cage and crosshead.

[0031] FIG. 4 is an isometric view of an embodiment of the crosshead **304** including a knuckle cage **400**. In this example, the crosshead **304** includes an interior portion **402** having a slot **404** that receives the knuckle cage **400**. The slot **404** is arranged between a first end **406** and a second end **408** of a body portion **410** of the crosshead **304**, which may act as a platform, as will be described herein. In other words, the slot **404** is recessed within the body portion **410** and/or may be represented as an indentation or a recessed area, thereby forming an area to receive the knuckle cage **400**.

[0032] The crosshead **304** further includes a radial hole **412**. In this example, the radial hole **412** is substantially aligned with the knuckle cage **400** and in operation the knuckle cage **400** may be installed through the radial hole **412**. Further illustrated is the aperture **306**, which may be utilized to provide fluids, such as lubrication, to the interior portion **402** of the crosshead **304**. For example, one or more flow passages (not pictured) may be cross drilled and/or routed within the body portion **410** to provide lubrication to predetermined locations.

[0033] Returning to the knuckle cage **400**, a cage hole **414** extends through a knuckle body **416** and is aligned with the radial hole **412**. In operation, a knuckle, described below, may extend through both the cage hole **414** and the radial hole **412**, which may eliminate the pins described above with respect to prior art configurations. The illustrated knuckle cage **400** further includes a cut out **418**, which removes a portion of the knuckle body **416** to provide access to a passage **420** formed by the knuckle hole **414**. In this example, the cut out **418** is substantially rectangular, but it should be appreciated that other shapes may be utilized for the cut out **418**. In one or more embodiments, the cut out **418** may be considered as a material portion removed from a

tubular or cylindrical component to provide access to an interior portion of the tubular or cylindrical component while still maintaining the general shape. In the embodiment of FIG. 4, the cut out **418** does not extend along an entire length **422** of the knuckle cage **400**. It should be appreciated that the cut out **418** may extend along the length **422** and that, in other embodiments, one or more additional closure features may be utilized to block access to the interior portion **402** in the absence of the knuckle body **416**. For example, in one or more embodiments, the cut out **418** may extend along the entire length **422** and the body portion **410** may include one or more features along the interior portion to block or otherwise restrict movement of components extending through the knuckle cage **400**. In other words, portions of knuckle cage **400** illustrated proximate the cage hole **414** may be removed and replaced with one or more inserts or portions coupled to the body portion **410**.

[0034] In this example, the knuckle cage **400** further includes recesses **424** formed at opposite ends that facilitate securing the knuckle cage **400** to the crosshead **304**. In this example, lock washers **426** are secured to the inner portion **402** (e.g., at the ends **406**, **408** on the platform portion) and overlap, at least partially, the recesses **424**, thereby blocking movement of the knuckle cage **400**. In various embodiments, movement of the knuckle cage **400** is blocked in multiple directions. For example, movement along the passage **420** is blocked by contacting the body **416** on each side of the recess. Furthermore, upward movement (e.g., perpendicular to the passage **420**) is blocked by both the overlap at the radial hole **412** and by the lock washers **426**. It should be appreciated that the recesses **424** are shown for illustrative purposes as one embodiment for securing the knuckle cage **400** within the crosshead **304**. In various other embodiments, the recesses **424** may be omitted in favor of other methods for securing the knuckle cage **400**, such as fasteners, press fittings, locking components, and the like. Furthermore, in at least one embodiment, the recesses **424** may have a shorter lateral length such that the recesses **424** do not extend substantially across the length **422**. That is, the recesses **424** may have an opening or width particularly selected to receive one or more fasteners, such as the lock washers **426**, to further block movement along the passage **420**. Furthermore, in certain embodiments, the lock washers **426** may be applied along the cutout **418** without the recesses **424**.

[0035] The illustrated example further includes a fluid passage **428** formed in the knuckle body **416** at the passage **420**. That is, the fluid passage **428** enables fluid flow into the passage **420**. In various embodiments, the fluid passage **428** may be described as being arranged substantially perpendicular to the cage hole **414**. As noted above, fluids such as lubrication may be directed via the apertures **306** and into the inner portion **402**, for example to provide lubrication along a pivot surface formed within the passage **420**. It should be appreciated that there may be more than one fluid passage **428** and that the fluid passages **428** may be particularly arranged at different locations.

[0036] In operation, a knuckle is installed within the passage **420** and may pivot about an axis, responsive to movement of the connecting arm. The illustrated knuckle cage **400** includes a pivot surface **430** along the passage **420**, which may be machined or otherwise prepared for contact with the knuckle. As will be described, lubricating fluids may form a hydrodynamic cushion between the knuckle

cage 400 and the knuckle. In various embodiments, the knuckle cage 400 may be formed or prepared to resist shock or other damages, for example by hardening or various other surface preparation methods. In the illustrated embodiment, the pivot surface 430 includes grooves or cutouts. In this example, both axial grooves 432 and lateral grooves 434 are illustrated, but it should be appreciated that the pivot surface 430 may be substantially smooth or may include more or fewer grooves 432, 434. In one or more embodiments, the grooves 432, 434 may receive lubricating fluid, which may reduce friction and provide additional hydrodynamic cushion to one or more components. In at least one embodiment, the grooves 432, 434 may be adjusted based upon the supplied lubrication. That is, the grooves 432, 434 may be longer/shorter, wider/thinner, deeper/shallower, and/or the like based, at least in part, on properties of the supplied lubrication.

[0037] FIG. 5 is an isometric view of the knuckle cage 400. As noted above, the knuckle body 416 is substantially cylindrical and includes the passage 420 extending there-through and cage holes 414 at each end. Further illustrated is the cut out 418 providing access to the passage 420. This example also includes a rectangular cut out 418, but it should be appreciated that other shapes may be utilized. As noted above, the cut out 418 does not extend the length 422 in this example, thereby leaving portions of the body 416 at each end. However, other embodiments may include cut outs 418 that extend the length 422. For example, in at least one embodiment, the portions of the body 416 may include one or more slots of passages coupled to the cut out 418, which may extend the cut out 418 along the length 422.

[0038] The knuckle cage 400 also includes the recesses 424 described above, but which are illustrated as ends or substantially planar ends of the knuckle cage 400. As shown, the recesses 424 have a depth 502 (taken from a top of the knuckle cage 400). Barriers 504 are formed at each end of the recesses 424 via the cut out 418, which may block movement of the knuckle cage 400 when secured within the crosshead 304. That is, in one or more embodiments, lateral movement parallel to the passage 420 may be blocked due to engagement between the lock washers 426 (FIG. 4) and the barriers 504. It should be appreciated that the lock washers described above for this configuration are for illustrative purposes and that various other methods may be utilized to secure the knuckle cage 400 to the crosshead 304. For example, apertures may be formed to receive fasteners. Additionally, clamps or the like may be utilized. Furthermore, other secure fittings, such as J-hooks and the like may be incorporated into various embodiments. Additionally, in one or more embodiments, the body portion 410 (FIG. 4) may include one or more extensions that may engage the recesses 424, among other options.

[0039] FIG. 6 is an isometric view of an embodiment of a connecting assembly 600 that includes the connecting rod 302 and a knuckle 602. In operation, the connecting rod 302 is secured to apertures (not shown) formed in the knuckle 602 via axial fasteners 604. In this example, the axial fasteners 604 are threaded fasteners, such as bolts, but it should be appreciated that other fasteners may be utilized in various embodiments. By way of example only, additional axial fasteners may include spring-loaded pins, rivets, cotter pins, clamps, and the like. Furthermore, it should be appreciated that the fasteners may extend through the knuckle 602 and engage apertures formed in the connecting rod 302.

Accordingly, various different fastening configurations may be utilized within the scope of the present disclosure. For example, pinning from the side, dovetailing, and the like may be utilized to affix the connecting rod 302 to the knuckle 602. Accordingly, separation on the retract stroke is prevented.

[0040] In this example, the knuckle 602 includes a cylindrical body 606 that includes a knuckle recess 608 that receives a mating end 610 of the connecting rod 302. As shown, both the mating end 610 and a mounting surface 612 within the recess 608 are substantially flat (e.g., planar). As noted above, planar coupling surfaces may facilitate improved alignment for fastening and also force distribution, which may improve the reliability and longevity of the components.

[0041] The recess 608 extends for a recess length 614 that is less than a knuckle length 616. However, it should be appreciated that the recess 608 may also be substantially equal to the recess length 614. Accordingly, respective edges 618 (e.g., a first edge 618A and a second edge 618B) are formed at each end of the recess 608, which may be utilized to facilitate alignment of the mating end 610. Furthermore, the edges 618 may further restrict movement of the mating end 610. In other instances, the recesses length 614 may extend the length of the knuckle length 616. It should be appreciated that other elements may also be used, or be used in place of the edges 618, to facilitate alignment of the mating end 610 with the mounting surface 612. For example, pins 620 (e.g., dowel pins) may be installed in the mating end 610, the mounting surface 612, or both to facilitate alignment. For example, the illustrated pins 620 may engage holes or openings formed in the mating surface 612. The pins 620 may be used for alignment and also to restrict movement. Furthermore, in one or more embodiments, the mating end 610 may include extensions or edges to replace the edges 618 of the knuckle 602. By way of example, the knuckle 602 may include one or more slots to receive the extensions of the mating end 610 to facilitate alignment between the components.

[0042] In this example, the knuckle 602 includes a fluid aperture 622 that may facilitate supplying lubrication to one or more components within the crosshead 304, as described above. For example, the fluid aperture 622 may align with the aperture 306 and/or the fluid passage 428 to permit fluids, such as lubricants, into the inner portion 402 of the crosshead 304.

[0043] As will be described below, in operation the knuckle 602 may be installed within the passage 420 of the knuckle cage 400. An outer surface 624 of the knuckle 602 may rotate along the passage 420 about an axis 626. Such rotation may be supplied via movement of the connecting rod 302, which as noted above may be coupled to the powered crankshaft 104. Accordingly, the pivoting movement of the knuckle 602 within the knuckle cage 400 may facilitate translation of the rotational movement of the powered crankshaft 104 into axial movement of the plunger (not pictured), which is coupled to the crosshead 304 by the pony rod 120, as shown in FIG. 1.

[0044] FIG. 7 is an isometric view of an embodiment of a crosshead assembly 700 in which the knuckle 602 and the knuckle cage 400 are both installed within the crosshead 304. The connecting rod 302 has been omitted for clarity, but it should be appreciated that in operation the connecting rod 302 may couple to the illustrated mounting surface 612 via

the mating end 610. As noted above, various features may be utilized to align and/or couple the connecting rod 302 to the mounting surface 612. For example, mating apertures 702 are illustrated along the planar mounting surface 612. It should be appreciated that the mating apertures 702 may include threads to facilitate coupling to the axial fasteners 604 or may be smooth holes for insertion of pins 620, among other mating options such as the above-described side pinning, dovetailing, and the like. Furthermore, it should be appreciated that the mating apertures 702 may be formed in the mating end 610 and that the fasteners and/or pins may extend through the knuckle 602. Additionally, the pattern of the apertures 702 is shown for illustrative purposes and any reasonable pattern or number of apertures 702 may be utilized.

[0045] As shown, the axis 626 of the knuckle 602 is aligned with an axis 704 of the passage 420. In other words, the knuckle 602 and the passage 420 are axially aligned within the interior portion 402 of the crosshead 304. As shown, the recess 608 in the knuckle 602 positions the mounting surface 612 substantially aligned with the locking recesses 424. However, it should be appreciated that the mounting platform 612 may be axially lower than or axially higher than the locking recesses 424. Furthermore, in this example, the edges 618 are within the ends of the knuckle cage 400. In other words, the recess 608 and cut out 418 substantially overlap to provide access to the mounting surface 612.

[0046] In operation, the connecting rod 302 is coupled to the mounting surface 612, which blocks movement of the knuckle 602 along the axis 626, effectively securing the knuckle 602 within the cage 400 and the crosshead 304. That is, locking the knuckle 602 within the cage 400 secures the knuckle 602 within the crosshead 304 via the lock washers 426 that secure the cage 400 to the crosshead 304. It should be appreciated that the knuckle 602 may still rotate about the axis 626, for example along the outer surface 624 (FIG. 6), which may be particularly selected and/or formed to facilitate the rotation while also reducing the likelihood of surface damage. Moreover, in various embodiments, the knuckle cage 400 may also be machined and/or formed from materials to reduce the likelihood of surface damage.

[0047] The example shown in FIG. 7 further includes a lubrication port 706 shown on the mating surface 612. This illustrated port 706 includes a seal groove 708, which may receive a seal (not pictured). One or more passages (e.g., lubrication channels) may provide lubricating fluids to the mating surface 612, which may reduce friction between the mating surface 612 and the mating end 610. In embodiments, the fluid may be transported via the aperture 306. Furthermore, in embodiments, the lubricating fluids may be directed into a gap surrounding the knuckle 602 to provide lubricating fluids for the rotation of the outer surface 624 along the passage 420.

[0048] FIG. 8 is an isometric cross-sectional view of the knuckle 602 including lubrication channels 800 providing lubrication, such as a lubricating fluid, to the mounting surface 612 via the lubrication port 706. In this configuration, the lubrication channels 800 extend through the body 606 of the knuckle 602 from the fluid apertures 622. It should be appreciated that the configuration shown in FIG. 8 is for illustrative purposes only, and there may be more or less than two channels 800. Furthermore, the channels 800 may be arranged at any angle, may join at different locations,

and the like. Furthermore, the channels 800 may not be in plane with one another. In operation, lubrication will provide a hydrodynamic cushion to the knuckle 602 and the knuckle cage 400. It should be appreciated that, in various embodiments, lubrication may be supplied at 706 with an outlet at 622.

[0049] FIG. 9 is an isometric view of an embodiment of the connection assembly 600 illustrating an alternative or additive connection method. In this example, the illustrated mating end 610 is coupled to the knuckle 602 via retaining pins 900 extending radially inward along the axis 626. It should be appreciated that these pins 900 may be used in place of the fasteners 604 or in addition to the fasteners 604. In at least one embodiment, mating end 610 includes pin holes 902 that align with knuckle pin holes 904 when the mating end 610 is positioned within the recess 608. The pins 900 may then extend through the aligned holes 902, 904 to block separation of the mating end 610 from the knuckle 602. It should be appreciated that the alignment mechanisms, described above, may also be used to facilitate joining the mating end 610 to the knuckle 602. For example, the mating end 610 includes mating apertures 906, similar to the mating apertures 702 of the mounting surface 612 in FIG. 7, that may receive one or more rods or pins to align the components together. Accordingly, in operation, the pins 900 may be arranged to resist a shear force.

[0050] FIG. 10 is a flow chart of an embodiment of a method 1000 for coupling a connecting rod to a crosshead. It should be appreciated that for this method, and any other method described herein, that there may be more or fewer steps. Furthermore, the steps may be performed in a different order, or in parallel, unless otherwise specifically stated. In this example, a knuckle cage is positioned within a crosshead 1002. For example, the knuckle cage may be aligned with an opening formed in the crosshead and then installed within an interior portion of the crosshead such as along a passage extending between first and second ends of the crosshead. The knuckle may be secured to the crosshead in one or more embodiments, such as by using one or more fasteners or other coupling methods, such as a press fit, an interference fit, or the like. In one or more embodiments, installation of the knuckle cage may further include aligning one or more components, such as lubrication passages, openings, and the like.

[0051] In various embodiments, the knuckle is installed within the knuckle cage 1004. In at least one embodiment, the knuckle is installed in a manner that enables rotation about an axis. That is, in various embodiments, the knuckle cage may serve as a bearing or journal for the knuckle to rotate, such as along a surface of the knuckle cage. In at least one embodiment, the knuckle is axially aligned with the knuckle cage 1004. Furthermore, in at least one embodiment, the knuckle is installed to facilitate alignment of one or more components, such as aligning a knuckle mounting surface with an opening or accessible portion of the knuckle cage and/or crosshead.

[0052] In at least one embodiment, a mating end of a connecting rod is aligned with a mounting surface of the knuckle 1006. Alignment may include positioning one or more pins or extensions to mate or extend into mating apertures or grooves. In at least one embodiment, alignment may include positioning the connecting rod between edges that define an opening or recess in the knuckle. The connecting rod may be coupled to the knuckle 1008. In at least

one embodiment, the coupling is an axial coupling. Axial coupling may include a coupling that is aligned perpendicular to a rotational axis of the knuckle. In at least one embodiment, axial coupling may include one or more threaded fittings, one or more pins, one or more claims, or various other coupling mechanisms that facilitate axial coupling. In at least one embodiment, coupling is performed radially or parallel to the rotational axis of the knuckle. For example, one or more pins may extend through the knuckle to mate with one or more apertures formed in the connecting rod. In at least one embodiment, both axial and radial coupling is utilized to secure the coupling rod to the knuckle. It should be appreciated that additional components may also be utilized and considered, such as alignment of flow passages for lubrication, which may lead to further adjustments or movement of various components of the system. In this manner, the connecting rod may be coupled to the knuckle.

[0053] This application is a continuation of U.S. Non-Provisional application Ser. No. 17/458,754, filed Aug. 27, 2021, titled “CONNECTING ROD ASSEMBLY FOR RECIPROCATING PUMP,” which claims the benefit of U.S. Provisional Patent Application No. 63/071,137, filed Aug. 27, 2020, titled “CONNECTING ROD ASSEMBLY FOR RECIPROCATING PUMP,” the full disclosures of which are hereby incorporated herein in their entireties by reference.

[0054] The foregoing disclosure and description of the disclosed embodiments is illustrative and explanatory of the embodiments of the disclosure. Various changes in the details of the illustrated embodiments can be made within the scope of the appended claims without departing from the true spirit of the disclosure. The embodiments of the present disclosure should only be limited by the following claims and their legal equivalents.

1. A connecting rod assembly, comprising:
 - a crosshead to connect a rod to a plunger;
 - a knuckle cage positioned within the crosshead, the knuckle cage including a passage;
 - a knuckle positioned within the passage of the knuckle cage and connected to the crosshead, the knuckle having a mounting surface, the mounting surface including a mating end to connect to the rod; and
 one or more lock washers positioned to block movement of the knuckle cage in a direction transverse to a rotational axis of the knuckle.
2. A connecting rod assembly comprising:
 - a crosshead to connect a rod to a plunger, the crosshead having one or more holes extending through the crosshead;
 - a knuckle cage positioned within the crosshead to extend through the one or more holes of the crosshead, the knuckle cage including a passage and a cutout;
 - a knuckle positioned within the passage of the knuckle cage, the knuckle having a mounting surface, the mounting surface including a mating end to connect to the rod;
 - a lubrication passage extending between an outer portion of the knuckle cage and the mounting surface;
 - a first lubrication aperture formed in the crosshead; and
 - a second lubrication aperture positioned in the knuckle cage, the knuckle cage being installed within the one or more holes such that the first lubrication aperture aligns with the second lubrication aperture and the second

lubrication aperture directs a fluid toward a rotational surface of the knuckle cage.

3. The connecting rod assembly of claim 2, further comprising:

- a third lubrication aperture positioned in the knuckle, the third lubrication aperture fluidly connected to a rotational surface of the knuckle; and

- one or more channels extending through the knuckle, the one or more channels positioned to direct a fluid to a lubrication port on the mounting surface.

4. A connecting rod assembly, comprising:

- a crosshead to connect a rod to a plunger;

- a knuckle cage positioned within the crosshead, the knuckle cage including a passage; and

- a knuckle positioned within the passage of the knuckle cage, the knuckle having a mounting surface, the mounting surface including a mating end to connect to the connecting rod at the mating end, at least a portion of the knuckle comprising a hardened material selected to resist wear.

5. A connecting rod assembly, comprising:

- a crosshead having an interior portion, a hole extending from a first end to a second end, and a pair of platforms positioned on opposite first and second sides of the hole;

- a knuckle cage positioned within the hole and extending from the first end to the second end and having a cage axis extending transverse to the pair of platforms, a pivot surface, and a recess positioned in the knuckle cage and extending through a portion of the knuckle cage, thereby to provide access to the pivot surface of the knuckle cage from a direction transverse to the cage axis, the pivot surface at least partially defining at least a portion of a cage passage extending along the cage axis; and

- a knuckle positioned within the cage passage, the knuckle axially aligned with the cage axis such that the knuckle, responsive to a force, rotates about the cage passage during operation.

6. The connecting rod assembly of claim 5, further comprising:

- one or more receiving apertures positioned in the mating surface, the one or more receiving apertures positioned to receive mating fasteners from a connecting rod to couple the connecting rod to the knuckle.

7. The connecting rod assembly of claim 5, further comprising:

- one or more receiving apertures positioned in the knuckle, the one or more receiving apertures positioned transverse to the cage axis, and wherein the one or more receiving apertures receive one or more pins to connect a connecting rod to the knuckle.

8. The connecting rod assembly of claim 5, further comprising:

- a pair of knuckle cage couplers secured to respective platforms on the first and second sides of the hole, each of the knuckle cage couplers extending, at least partially, over the knuckle cage to block axial movement of the knuckle cage in at least two directions.

9. The connecting rod assembly of claim 5, wherein a lubrication passage provides a lubricating fluid along the pivot surface of the knuckle cage.

10. The connecting rod assembly of claim **5**, wherein the recess positioned in the knuckle cage has a recess length that is less than a knuckle cage length.

11. The connecting rod assembly of claim **5**, further comprising:

one or more alignment apertures positioned in the knuckle, the one or more alignment apertures positioned transverse to the cage axis, and wherein the one or more alignment apertures receive one or more pins associated with the knuckle.

12. The connecting rod assembly of claim **5**, further comprising:

one or more fasteners extending from the knuckle, the one or more fasteners positioned perpendicular to the cage axis, and wherein the one or more fasteners engage one or more apertures positioned in a connecting rod to couple the connecting rod to the knuckle.

13. A method comprising:

positioning a knuckle cage within a crosshead;

positioning a knuckle within the knuckle cage;

aligning a mating end of a rod with a mounting surface of the knuckle;

connecting the rod to the knuckle; and

aligning a first flow passage extending through the crosshead with a second flow passage extending through the knuckle cage, to thereby provide a lubricating fluid through the first flow passage and the second flow passage during operation.

* * * * *